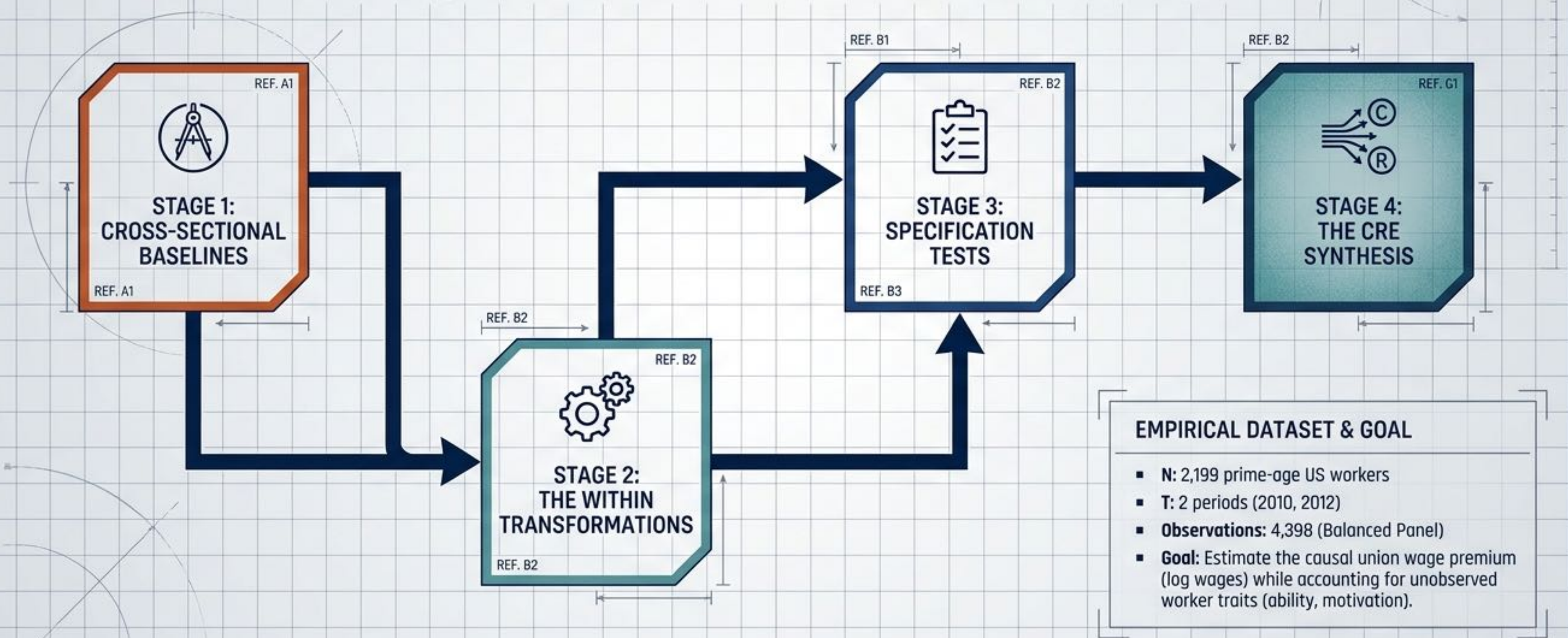


UNPACKING PANEL DATA METHODS: RESOLVING OMITTED-VARIABLE BIAS

A Methodological Guide from Pooled OLS to Correlated Random Effects (CRE) in Python.



The Core Tension: Between vs. Within Variation

The dominant signal: 1,805 workers are never unionized, 321 are always unionized. Cross-sectional estimators rely on this massive block of variance.

The narrow signal: Only 73 out of 2,199 workers (3.3%) switch union status between 2010 and 2012. Fixed Effects estimators must rely entirely on these 73 individuals.

93.9% Between Workers

**6.1%
Within
Workers**

The narrow signal: 1,805 workers are never unionized, 321 switch union status between 2010 and 2012. Fixed Effects estimators must rely entirely on these 73 individuals.

Because Fixed Effects discards 93.9% of the variation, its standard errors will mechanically inflate. The choice between estimators is a structural trade-off between bias and precision.

The Cross-Sectional Baselines: POLS & Between

THEORY

Pooled OLS (POLS): Treats all rows independently.
Between Estimator: Collapses data to worker means.

Analogy: Averaging the grades of a class without realizing some students took the exam twice. It ignores the panel structure.

IMPLEMENTATION (PYTHON)

```
fit_pols = pyfixest.feols("lwage ~ union",  
                          vcov="HC1")  
fit_betw = pyfixest.feols("lwage ~ union",  
                          data=df_between, vcov="HC1")
```

EMPIRICS & INTERPRETATION

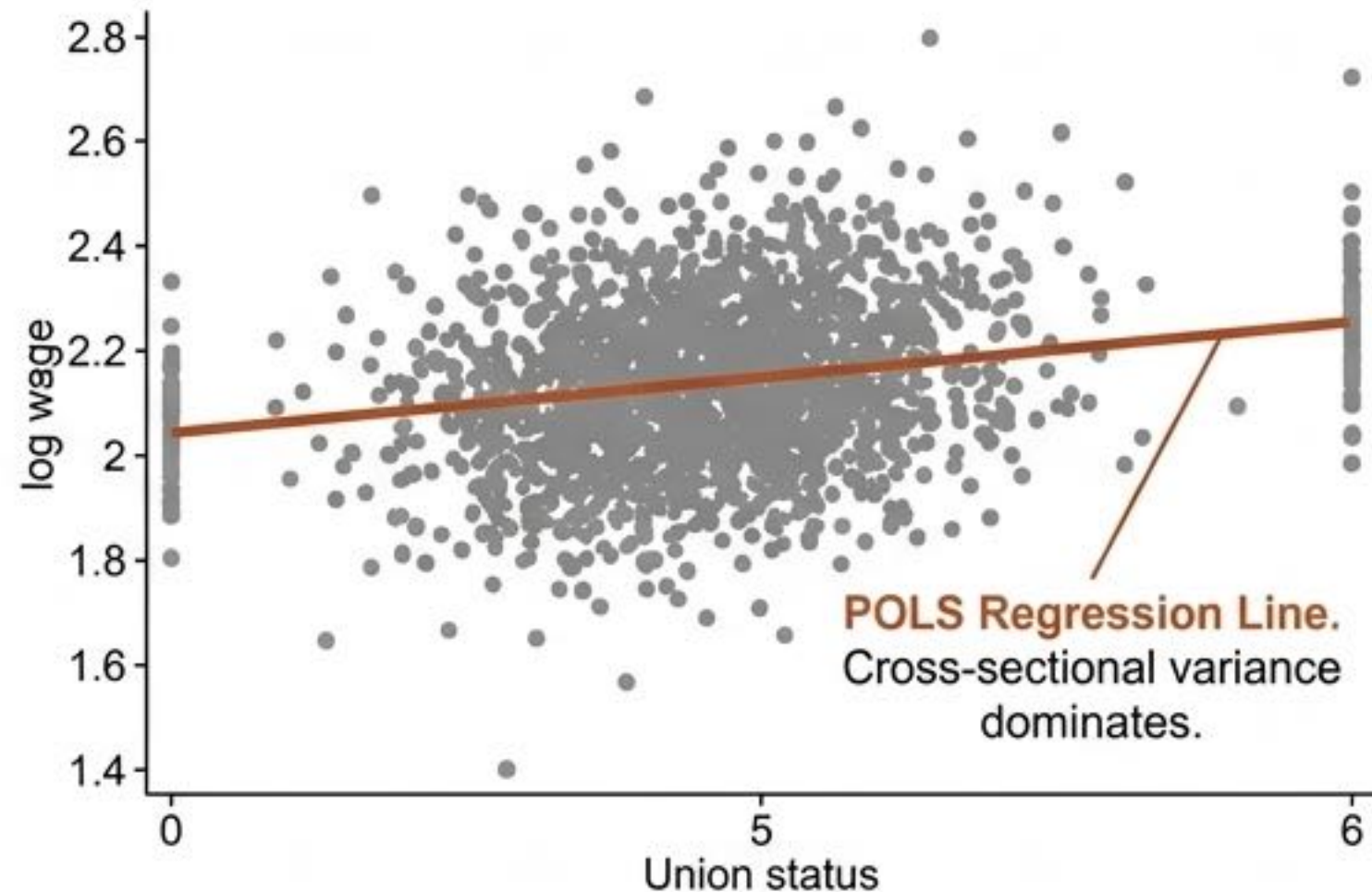
Method	Estimate	Standard Error
POLS	+0.0750	(0.0231)
Between	+0.0662	(0.0311)

Results context: Suggests a **~7.5 log point** union wage premium.

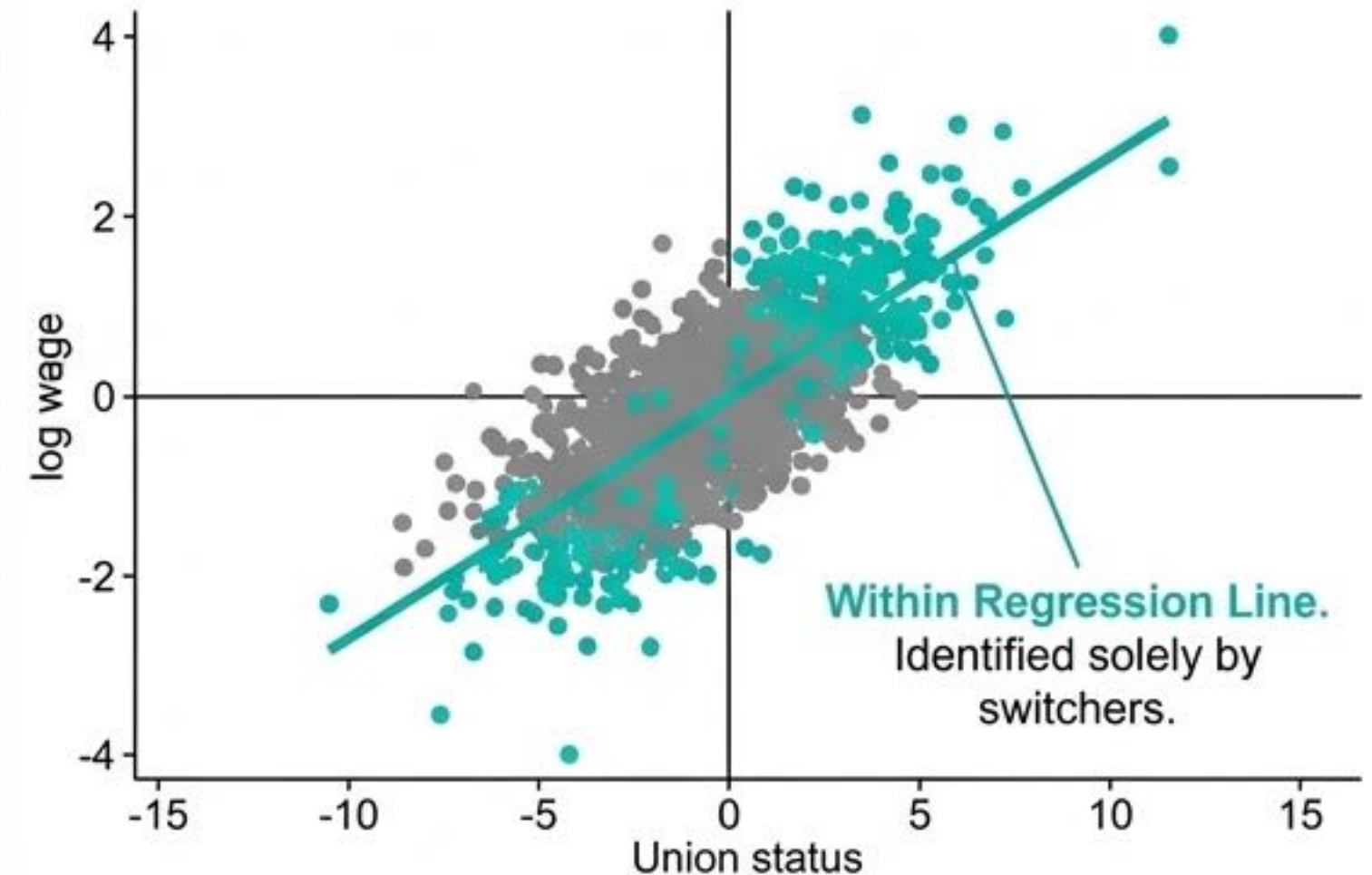
The Flaw (Omitted-Variable Bias): This mixes
this Flaw (Omitted-Variable Bias): This mixes the true union effect with fixed traits. If highly motivated workers systematically avoid unions, this +0.0750 confounds union membership with lower baseline earning ability.

Concept: The Within Transformation

Raw Data



Demeaned Data



The Demeaning Lens

Subtracting the watermark: Before demeaning, every page is dominated by a unit-specific watermark. The within transformation subtracts the time-series mean of each unit from its observations, leaving only the unique text underneath (the pure within-unit change).

Fixed Effects (FE & TWFE): Isolating the Within Signal

THEORY

- **Fixed Effects (FE):** Absorbs the unobserved unit effect to isolate the pure within-worker effect.
- **Two-Way Fixed Effects (TWFE):** Adds year effects to simultaneously absorb macroeconomic shocks.

IMPLEMENTATION (PYTHON)

One-Way FE: `pyfixest.feols("lwage ~ union | ID", vcov="HC1")`

TWFE: `pyfixest.feols("lwage ~ union | ID + year", vcov={"CRV1": "ID"})`

EMPIRICS & INTERPRETATION

Method	Estimate	Standard Error
One-Way FE	+0.2103	(0.0812)
TWFE	+0.2113	(0.0792)

Methodological Insight: The premium nearly triples from POLS (7.5 -> 21.0 log points). This dramatic shift implies negative selection: workers with unobserved traits that lower baseline wages are more likely to join unions, which suppressed the naive POLS estimate.

The Trade-off: Precision drops sharply. Standard errors inflate 3.4x because the model discards all non-switchers.

First Differences (FD) and the Equivalency Scale

Synthesizing the equivalence of FD and FE when $T=2$.

CONCEPT

First Differences subtracts last period from this period.

Analogy: Subtracting yesterday from today vs. subtracting the average day from today. With only two time time periods, differencing and demeaning are identical transformations of the exact same 73 switchers' data.

THE PROOF BOX: $T=2$ IDENTITIES

Identity 1: FD without an intercept = One-way FE

Both yield exactly **+0.2103**. The mathematics balance perfectly because with two periods, the demeaned values are exact halves of the difference.

Identity 2: FD with an intercept = Two-way FE (TWFE)

Both yield exactly **+0.2113**. The FD intercept (+0.0727) absorbs the exact same common 2010-2012 wage growth that the TWFE year effect absorbs.

The Efficiency Play: Random Effects (RE)

THEORY

- Treats unobserved worker traits as a random draw that is uncorrelated with regressors.
- **Analogy:** Trusting that worker-specific traits are just random noise, which allows us to safely blend both Between and Within data using GLS for maximum statistical power.

IMPLEMENTATION (PYTHON)

```
model =  
linearmodels.panel.RandomEffects(...)  
  
model.fit(cov_type="robust")
```

EMPIRICS & INTERPRETATION

RE Estimate:	+0.1092
RE Standard Error:	(0.0299)

The Catch: The standard error is vastly improved (2.7x smaller than FE), and the estimate lands suspiciously close to the naive POLS.

If the **no-correlation assumption fails** (i.e., if selection bias exists), this statistical efficiency is completely useless because the estimate is fundamentally biased.

Diagnostic 1: The Hausman Test & The Robustness Flaw

THE TEXTBOOK VERDICT

- **Methodology:** Tests if FE and RE estimates diverge significantly. If $p < 0.05$, we reject RE consistency.
- **Results:** $H = 5.62$, $p = 0.018$.
- **Textbook Conclusion:** Reject Random Effects, proceed with Fixed Effects.



THE METHODOLOGICAL WARNING

The Flaw: The classic Hausman formula rigorously requires **classical** (homoskedastic, uncorrelated) standard errors to be valid.

The Trap: Analysts often try to fix this by plugging **robust** standard errors into the classic formula. Doing so drops H to 1.79 ($p = 0.180$), failing to reject RE.

The Rule: This plug-in number has **no valid** chi-square distribution. When standard errors must be robust, the textbook Hausman test **must be abandoned** for the **Mundlak approach**.

Concept: The Mundlak Term (Correlated Random Effects)

Analogy: A Diplomatic Peace Treaty. CRE delivers the robust identification of FE and the **efficient framework** of RE in one unified regression.

$$y_{it} = \underbrace{\beta(x_{it})}_{\text{The Within Effect.}} + \underbrace{\gamma(\bar{x}_i)}_{\text{The Mundlak Term.}} + a_i + u_{it}$$

The Within Effect.
Numerically identical
to the FE coefficient.

The Mundlak Term. The unit mean of the regressor. Captures the difference between Between and Within estimators. Serves as a **direct test for selection bias**.

Frisch-Waugh-Lovell (FWL) Theorem Logic: By controlling for a worker's average union status across the panel, the model is mathematically forced to rely exclusively on within-variance for the main beta coefficient.

Implementation & Interpretation: Correlated Random Effects

THEORY

Adds the unit mean of time-varying regressors to a Random Effects framework.

Directly tests for correlation between unobserved traits and the regressor.

IMPLEMENTATION (PYTHON)

```
df['union_bar'] = df.groupby('ID')['union'].transform('mean')  
fit_cre = RandomEffects(df['lwage'],  
exog_cre).fit(cov_type='robust')
```

EMPIRICS & INTERPRETATION

Parameter	Estimate	p-value
Within Coef (β)	+0.2103	(Matches FE)
Mundlak Term (γ)	-0.1441	$p = 0.072$

Diagnostic Insight: The negative γ term mathematically equals the Between estimate minus the Within estimate. Its negative sign confirms negative selection into unions.

At $p=0.072$, evidence against RE is suggestive (not strictly conclusive at 5%), but this test remains statistically valid even with robust standard errors.

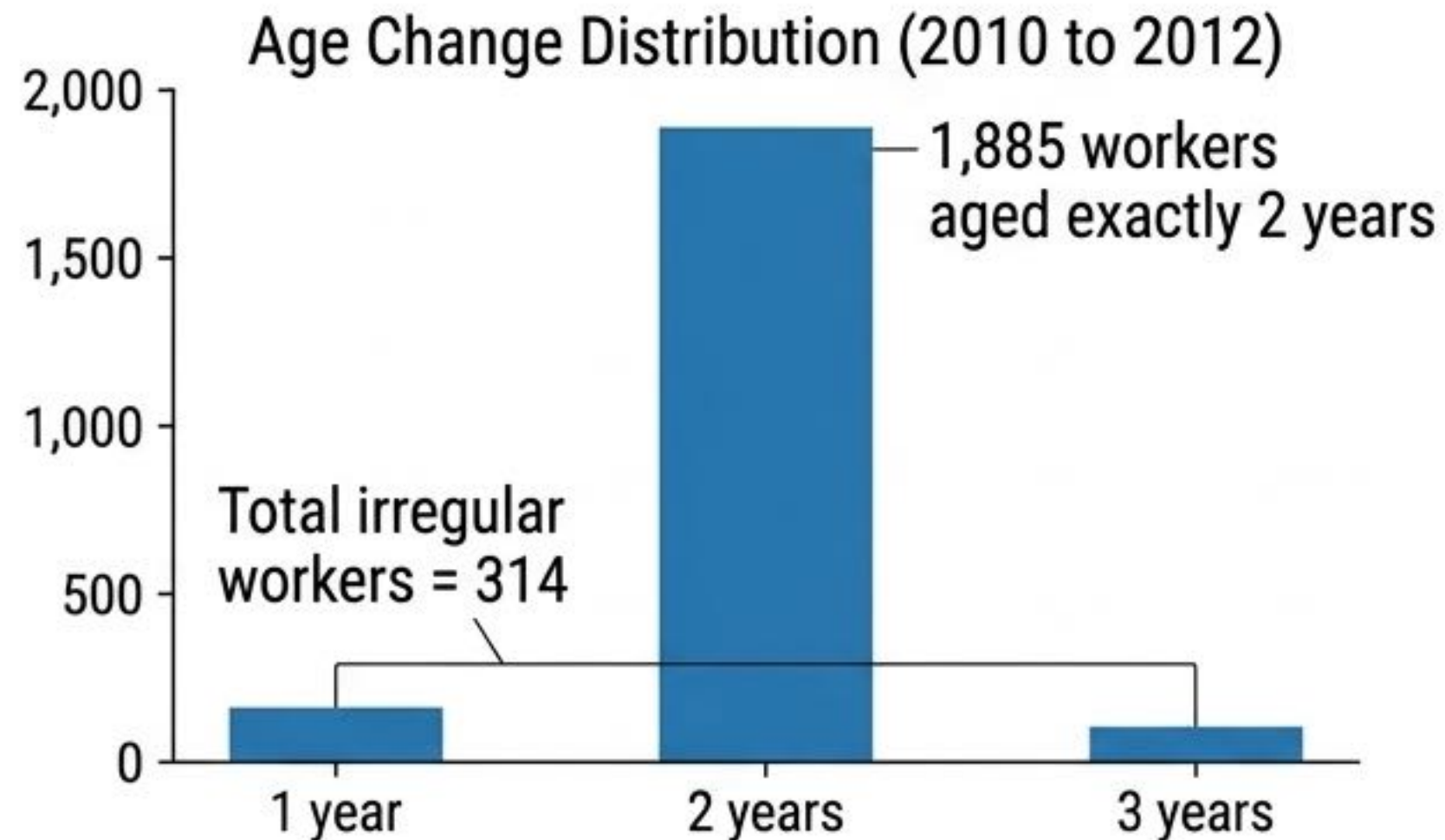
Adding Controls & The Age Anomaly

Expanding models with **covariates** (age, schooling, gender).

The Anomaly

In **POLS**, the **Age** coefficient is **+0.021** (Positive return to age).

In **TWFE & CRE**, the **Age** coefficient flips to **-0.058** (Negative return to age). Why?



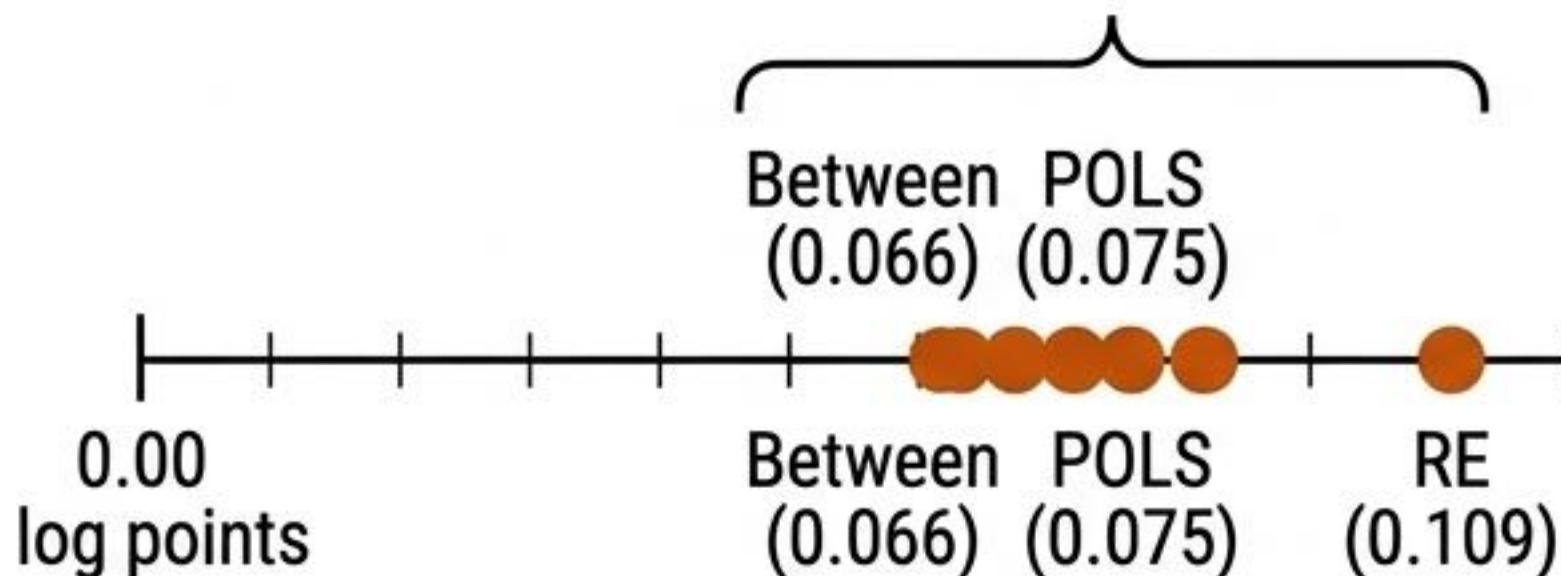
Because $T=2$, the Year Fixed Effect completely absorbs the mechanically advancing 2-year age gap. The negative -0.058 coefficient is therefore identified solely by the 314 irregularly interviewed workers.

Methodological Lesson: Always check what variation survives the within transformation. **Do not blindly interpret mechanically advancing variables.**

Synthesizing the Selection Story

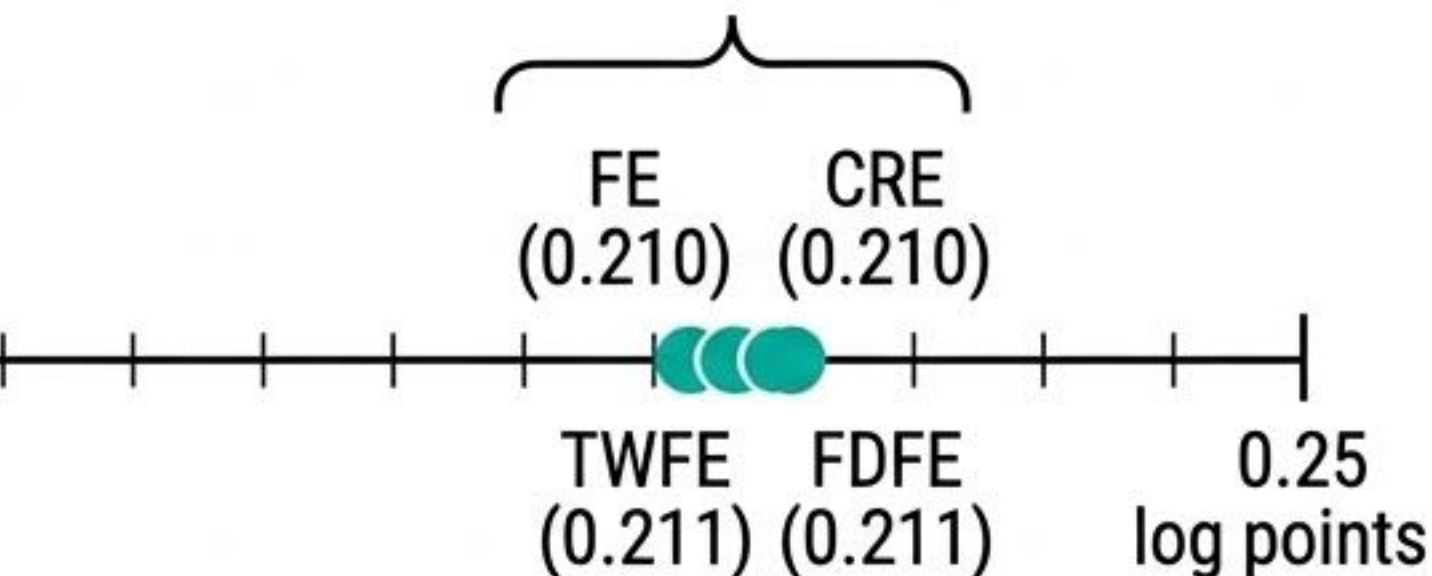
Econometrics is a lens for specific causal questions.

Cross-Sectional Lenses



Confounds the union effect with baseline earning ability.
Answers: How do union and non-union workers compare?

Within Lenses



Isolates pure state changes.
Answers: What happens when the exact same worker changes status?

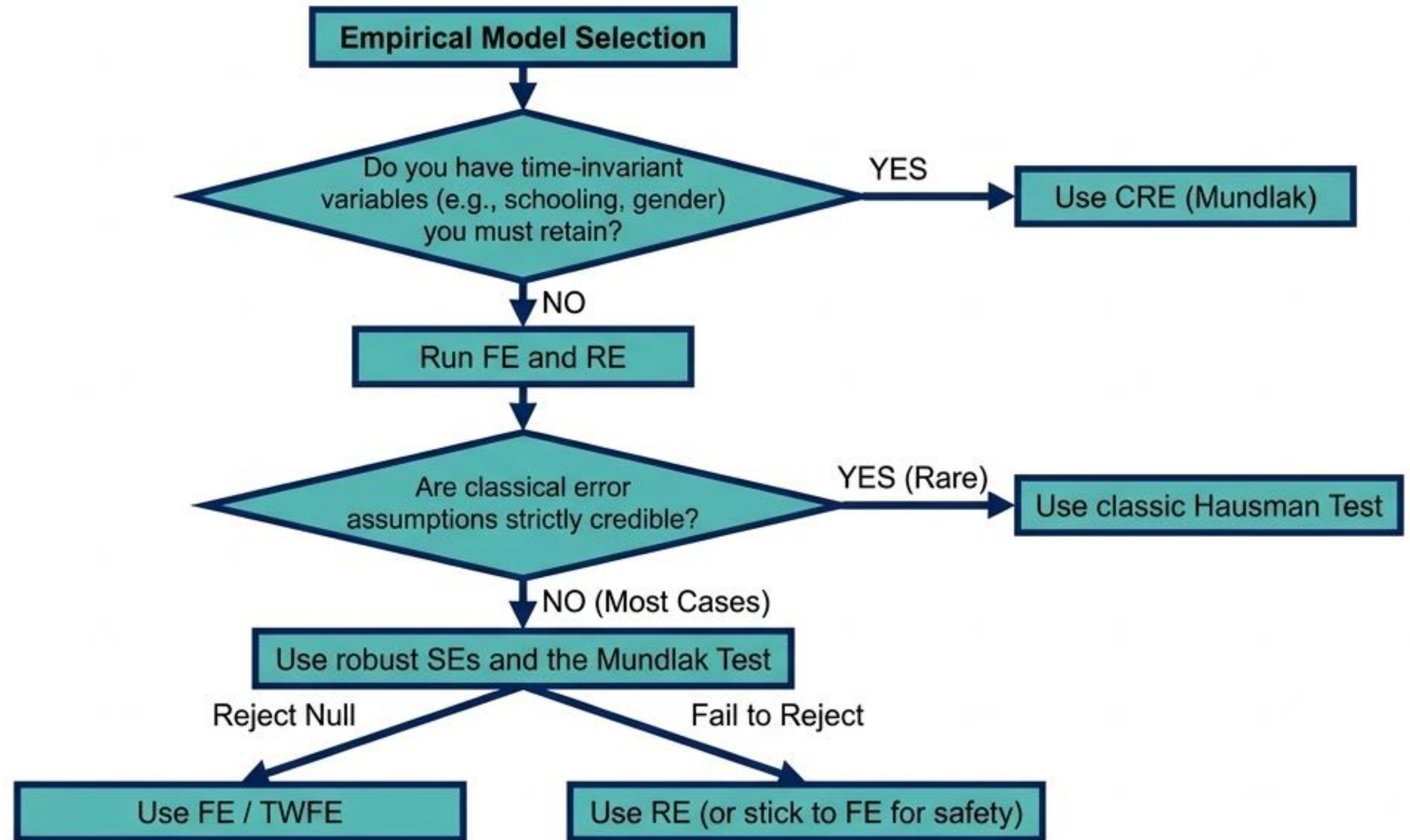
The massive gap between these clusters is not statistical noise—it is the empirical footprint of omitted variable bias (negative selection).

The Estimator Master Matrix

Method	Estimate (Log Points)	Standard Error	Variation Used & Key Assumption
POLS	0.075	0.023	All variation; ignores panel structure
Between	0.066	0.031	Cross-sectional means only
RE	0.109	0.030	GLS-weighted; Assumes unobserved traits are uncorrelated with x
FDFE	0.211	0.079	Within-worker differences
FE	0.210	0.081	Within-worker deviations from the mean
TWFE	0.211	0.079	Within-worker deviations net of year effects
CRE	0.210	0.070	RE with Mundlak terms (Extracts FE within)

Notice the 2.5x to 3.5x jump in standard errors for Within methods, reflecting the loss of cross-sectional variation.

The Specification Decision Tree



Correlated Random Effects (CRE) is increasingly the **modern default** because it bypasses the structural limits of one-way FE while maintaining robust identification.

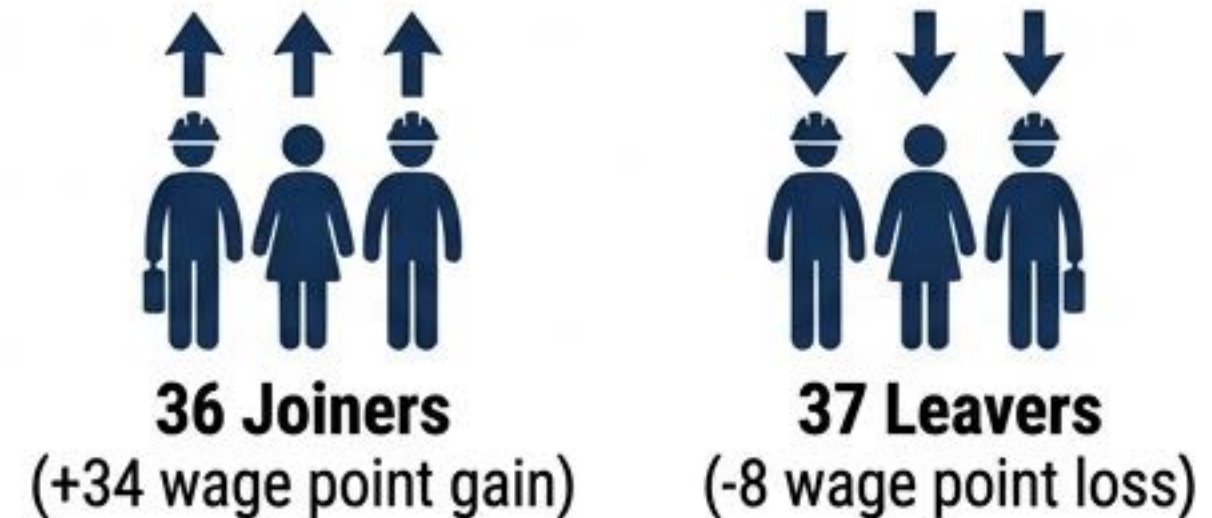
Conclusion & Empirical Best Practices

THE VERDICT

The CRE (Mundlak) specification should be the sensible default for modern panel data analysis. Why?

1. Reproduces the Fixed Effects within-coefficient strictly.
1. Reproduces the Fixed Effects within-coefficient strictly.
2. Retains time-invariant covariates (schooling, gender).
3. Provides a built-in, cluster-robust specification test.

THE CAUTIONARY CAVEAT



The robust 21 log point estimate relies exclusively on these 73 switchers over a tiny 2-year window. The effects are highly asymmetric.

When expanded to all 5 waves of the NLSY data, the TWFE estimate falls to 0.040.

Rule of Thumb: Never over-interpret robust estimators if the underlying identifying variation is extremely small or highly specific. Context is always required.